Operating Systems

Operating system

An operating system (OS) is system software that manages computer hardware and software resources and provides common services for computer programs, acting as an intermediary between the applications that a user runs and the physical hardware on which those applications execute.

Time-sharing operating systems schedule tasks for efficient use of the system and may also include accounting software for cost allocation of processor time, mass storage, peripherals and other resources, a design that emerged in the 1960s when computers were expensive and shared among many users.

Kernel

The kernel is the central component of an operating system; it is the first program loaded after the bootloader, it remains resident in memory, and it mediates all access to the processor, memory and input/output devices on behalf of application programs.

In a monolithic kernel, such as the one used by Linux, the core services including the scheduler, memory manager, file systems and device drivers all execute in a single privileged address space, whereas a microkernel moves most of these services into separate user-space processes that communicate by message passing.

Applications request services from the kernel through system calls, which switch the processor from user mode to kernel mode so that privileged instructions can be executed safely, and then return control to the application once the request has been serviced.

Process management and concurrency

A process is an instance of a program in execution, comprising the program code, its current activity as represented by the program counter and registers, a stack, a heap and the resources allocated to it, and each process is described by a data structure called the process control block.

Multitasking is achieved through context switching, in which the operating system saves the state of the currently running process and restores the saved state of another, giving the appearance that many processes run simultaneously on a single processor core.

When several processes share data, a race condition can occur if the final outcome depends on the order in which their instructions are interleaved; to prevent this, operating systems provide synchronisation primitives such as the semaphore, introduced by Edsger Dijkstra in 1965, and the mutex.

A deadlock is a situation in which each process in a set is waiting for a resource held by another process in the same set, and it can arise only when four conditions hold simultaneously: mutual exclusion, hold and wait, no preemption, and circular wait.

Memory management

Virtual memory is a memory-management technique that gives each process the illusion of a large, contiguous address space by mapping virtual addresses to physical addresses through page tables, allowing the system to use secondary storage as an extension of main memory.

When a process references a page that is not currently in physical memory, the hardware raises a page fault, and the operating system responds by loading the required page from disk, possibly evicting another page according to a replacement policy such as least recently used.

File systems

A file system organises data on storage devices into named files and hierarchical directories, records metadata such as size, ownership and permissions, and uses structures such as inodes or allocation tables to locate the blocks that make up each file.